

Let $P(x) = x^7 + a_6x^6 + \cdots + a_0$ be a good polynomial. The conditions are:

1. $a_i \in \mathbb{Z}$ and P is monic of degree 7.
2. $|a_i|$ is odd for all $i = 0, \dots, 7$ (with $a_7 = 1$). Note that $a_i \neq 0$.
3. $|a_i| < 2025^2$ for all i .
4. $\{|a_i| : i = 0, \dots, 7\} \subseteq \{\lambda : P(\lambda) = 0\}$.

Let $R = \{|a_i| : i = 0, \dots, 7\}$ be the set of the absolute values of the coefficients. Since $|a_7| = 1$, we have $1 \in R$. Let $u = |a_0|$. Since $a_0 \in R$ is not required (only absolute values), but $|a_0| \in R$ by definition, we have $u \in R$. From condition 4, every element $r \in R$ is a root of $P(x)$. Since $P(x)$ is a monic polynomial with integer coefficients, the product of its roots is $\pm a_0$. Let the roots be $\lambda_1, \dots, \lambda_7$. Then $\left| \prod_{j=1}^7 \lambda_j \right| = |a_0| = u$. Since $u \in R$, u is a root of $P(x)$. Let's say $\lambda_1 = u$ (since $u > 0$). Then $u \left| \prod_{j=2}^7 \lambda_j \right| = u$, which implies $\left| \prod_{j=2}^7 \lambda_j \right| = 1$. Also, for any $r \in R$, r is a root. If there exists $r \in R$ with $r > u$, then r contributes a factor r to the product of the moduli of the roots (since r is real and positive, or we consider its modulus). This would imply $u = \prod |\lambda_j| \geq r > u$, a contradiction. Thus, $u = \max(R)$.

We consider two cases based on the value of u .

Case 1: $u = 1$. Since $u = |a_0| = 1$ and u is the maximum of R , all $|a_i| \leq 1$. Since $|a_i|$ are odd integers, $|a_i| = 1$ for all i . Thus $a_i \in \{1, -1\}$. The condition $R \subseteq \text{Roots}(P)$ becomes $\{1\} \subseteq \text{Roots}(P)$, so $P(1) = 0$. $P(1) = 1 + a_6 + a_5 + \cdots + a_0 = 0$. Let k be the number of coefficients a_i (for $i = 0, \dots, 6$) equal to -1 . Then the number of 1's is $7 - k$. The sum is $1 + (7 - k)(1) + k(-1) = 1 + 7 - 2k = 8 - 2k$. Setting $8 - 2k = 0$ gives $k = 4$. The number of such polynomials is the number of ways to choose 4 coefficients to be -1 out of 7, which is $\binom{7}{4} = 35$.

Case 2: $u > 1$. Since u is odd, $u \geq 3$. Also $1 \in R$ and $u \in R$. We have $R = \{1, u\}$ because any element in R must be in the set of roots, and for the specific structure required by the coefficients, we will find $R \subseteq \{1, u\}$. The polynomial must have roots 1 and u . Thus, $P(x)$ is divisible by $(x - 1)(x - u) = x^2 - (u + 1)x + u$. Let $P(x) = (x^2 - (u + 1)x + u)Q_1(x)$, where $Q_1(x) = x^5 + c_4x^4 + c_3x^3 + c_2x^2 + c_1x + c_0$ is a monic polynomial with integer coefficients. The condition $|a_i| \in \{1, u\}$ for all coefficients a_i leads to constraints on c_i . Specifically:

1. $|a_1| = |u(c_1 - c_0) - c_0| \in \{1, u\}$ implies $c_1 = c_0$.
2. $|a_2| = |u(c_2 - c_0)| \in \{1, u\}$ implies $|c_2 - c_0| = 1$.
3. $|a_3|$ condition implies $c_3 = c_2$.
4. $|a_4|$ condition implies $|c_4 - c_2| = 1$.
5. $|a_5|$ condition implies $c_5 = c_4$.
6. $|a_6| = |c_4 - u - 1| \in \{1, u\}$.

Given $c_0 \in \{1, -1\}$, the choices for signs in the differences lead to a few paths for the coefficients c_i . The condition $|c_4 - c_2| = 1$ with $c_2 \in \{c_0 \pm 1\}$ means $c_4 \in \{c_2 \pm 1\}$. Calculating the possible values for c_4 based on the paths of choices for c_2 and c_3 relative to c_0 :

- We have $c_0 \in \{1, -1\}$.
- $c_2 \in \{c_0 + 1, c_0 - 1\}$.
- $c_4 \in \{c_2 + 1, c_2 - 1\}$. This results in $c_4 \in \{-3, -1, 1, 3\}$. The condition $|c_4 - u - 1| \leq u$ (since $|a_6| \in R \implies |a_6| \leq u$) implies $c_4 - u - 1 \geq -u \implies c_4 \geq 1$. So $c_4 \in \{1, 3\}$. If $c_4 = 1$, then $|a_6| = |1 - u - 1| = u$, which is valid for any u . If $c_4 = 3$, then $|a_6| = |3 - u - 1| = |2 - u| = u - 2$ (for $u \geq 2$). We need $u - 2 \in \{1, u\}$. $u - 2 = 1 \implies u = 3$. $u - 2 = u \implies -2 = 0$ (impossible). So $c_4 = 3$ is only possible if $u = 3$.

Counting the valid polynomials:

- For a fixed u , c_0 has 2 choices.
- c_2 is determined by c_0 and a sign (2 choices).
- c_4 is determined by c_2 and a sign (2 choices). Total paths for (c_0, c_2, c_4) signs is $2 \times 2 \times 2 = 8$.
Let's check which paths yield valid c_4 : Paths lead to $c_4 \in \{-3, -1, 1, 3\}$.
- $c_4 = 1$ is valid for all $u \geq 3$. The number of paths leading to $c_4 = 1$ is 3.
- $c_4 = 3$ is valid only for $u = 3$. The number of paths leading to $c_4 = 3$ is 1. (Paths leading to $c_4 \in \{-3, -1\}$ are invalid as they violate $|a_6| \leq u$).

So, for any odd integer $u \geq 3$ with $u < 2025^2$:

- If $u = 3$, there are $3 + 1 = 4$ valid polynomials.
- If $u \geq 5$, there are 3 valid polynomials.

We need to sum over all valid u . u is an odd integer such that $3 \leq u < 2025^2$. Let $M = 2025^2 = 4100625$. The possible values for u are $3, 5, \dots, M - 2$. The number of such values is $\frac{(M-2)-3}{2} + 1 = \frac{M-3}{2}$. Number of values ≥ 5 is $\frac{M-3}{2} - 1 = \frac{M-5}{2}$. For these values, there are 3 polynomials each. For $u = 3$, there are 4 polynomials.

Total count for $u > 1$: $4 \times 1 + 3 \times \frac{M-5}{2} = 4 + 3(2050310) = 4 + 6150930 = 6150934$.

Total good polynomials = (Case 1) + (Case 2) = $35 + 6150934 = 6150969$.